

# Rastermap

## Clustering: `n_clusters=None` vs specified values

- `n_clusters=None` or 0: disables clustering, sorts individual neurons. This is only feasible for small datasets ( $\sim < 200$  neurons), as the sort becomes NP-hard for large  $N$ . Useful for fine-grained neuron sorting.
- `n_clusters=N` (e.g. 50–200): applies scaled k-means before sorting. Each cluster is treated as a “superneuron”. Default is 100. More clusters preserve detail but slow down sorting; fewer clusters compress more but risk mixing signals.

### Example:

- 54 neurons  $\rightarrow$  `n_clusters=None`
  - Whole-brain recording (thousands)  $\rightarrow$  `n_clusters=100`
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## Clustering Step Details

1. **PCA** on (neurons  $\times$  time) to extract `n_PCs=200` components  $\rightarrow$  spatial basis.
2. **Clustering** in PCA space using scaled k-means  $\rightarrow$  `U_nodes` (cluster centroids), `X_nodes` (average trace).
3. **Temporal basis** comes from the cluster traces (`X_nodes`).
4. If `n_clusters=None`: each neuron becomes its own cluster.

This step compresses data from  $N$  neurons  $\rightarrow$   $k$  clusters and denoises by averaging.

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## Superneurons and Binning

- A **superneuron** is the average of a group of similar neurons.
  - After sorting, neurons can be binned into superneurons for display (e.g. 50-neuron bins).
  - Binning is distinct from clustering: it’s for visualization.
  - Smaller clusters than bin size are fine; bins may cross cluster boundaries.
  - Superneurons smooth out noise and reduce display rows.
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## Sorting via TSP + Segment Shifting

- Rastermap orders clusters using a variant of the Traveling Salesman Problem (TSP).
- Steps:

1. Compute similarity matrix ( $B B^T$ ) from cluster traces.
2. Define a target matrix balancing global similarity and local continuity.
3. Use heuristic segment shifting to optimize order (NP-hard problem).
4. `verbose_sorting=True` shows shift steps.

**Locality parameter  $w$**  controls: -  $w=0$ : prioritize global groupings. -  $w=1$ : prioritize local sequences.

**Time lag:** allows asymmetric similarity (lead/lag in activity).

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## Embedding and Upsampling

- After cluster ordering, neurons get assigned positions in 1D.
- `grid_upsample=10` creates more refined positions between clusters.
- Each neuron gets a fractional coordinate (**embedding**), not just a cluster label.
- Enables intra-cluster ordering based on small differences.

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## Final Outputs

- **embedding:** fractional 1D coordinate per neuron.
- **embedding\_clust:** cluster ID per neuron.
- **isort:** final sorted index list.
- Sorted similarity matrix is approximately block-diagonal.

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## Summary: Model Step by Step

| Step                                  | Compression | Modeling Purpose                              |
|---------------------------------------|-------------|-----------------------------------------------|
| PCA                                   | Yes         | Reduces time dimensionality, extracts signals |
| Clustering $\rightarrow$ superneurons | Yes         | Forms prototypes, reduces problem size        |
| TSP sorting                           | No          | Finds 1D manifold of similarity               |
| Segment shifting                      | No          | Optimizes the TSP ordering                    |
| Cluster upsampling                    | Kind of     | Fine-grains neuron placement                  |
| Neuron Binning                        | Yes         | Smooths output for display                    |

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## References

- Rastermap: a discovery method for neural population recordings (Nature Neuroscience, 2024)

- [GitHub - MouseLand/rastermap](#)
- Issues: [#36](#), [#38](#)